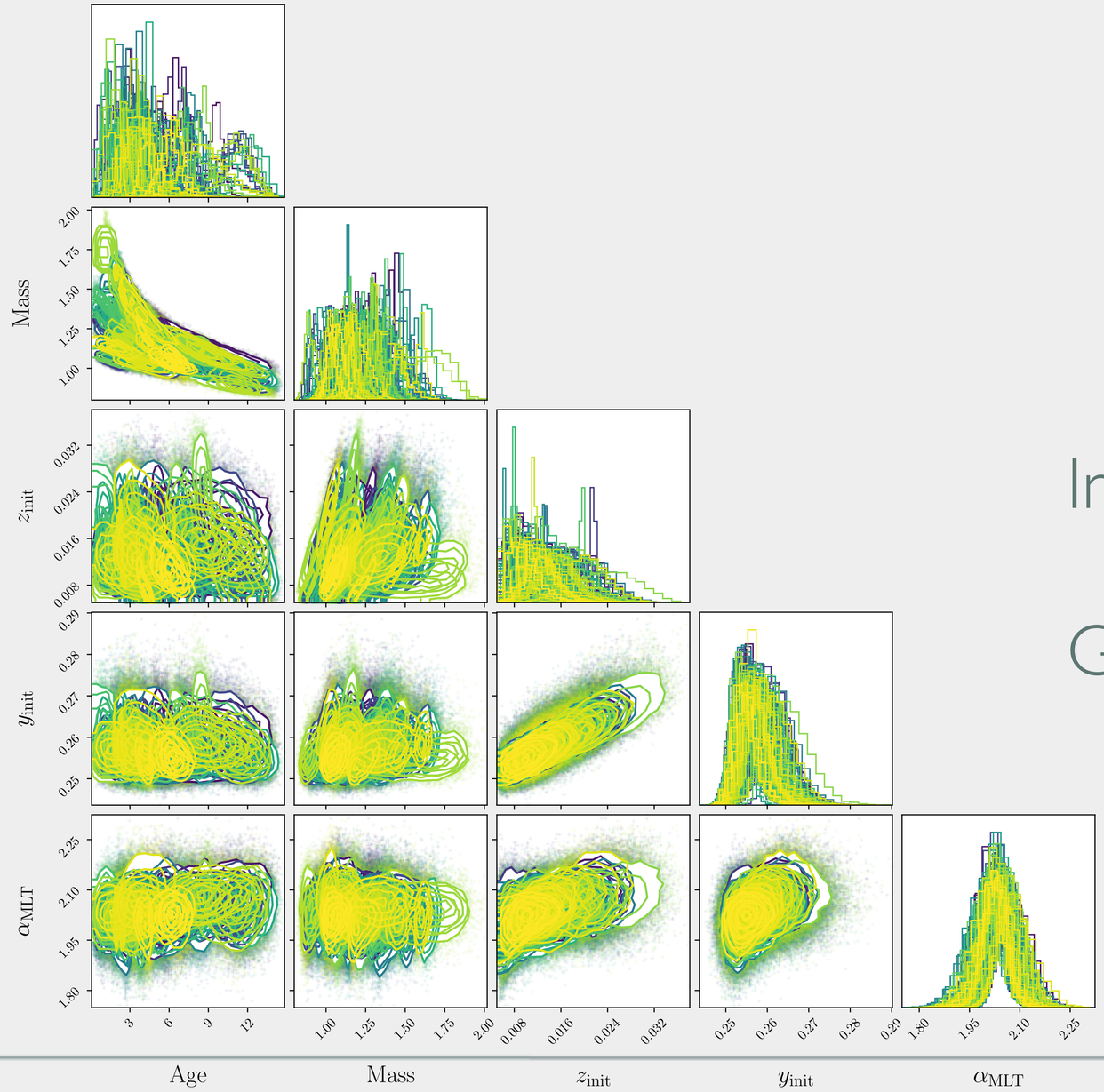
Inference from Scientific Data



Introduction Lecture

Guy Davies

Inference - Introduction

About the course:

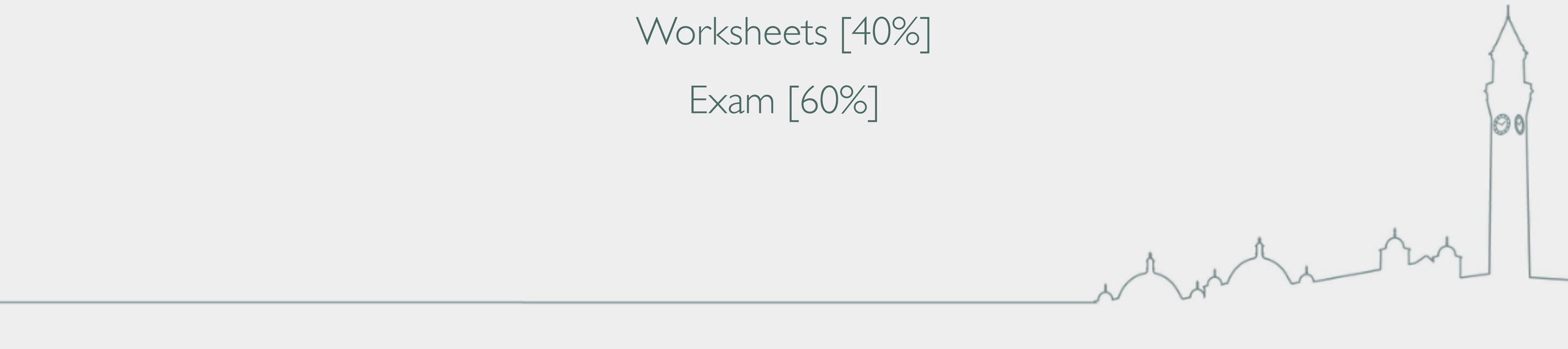
Monday 14:00 ARTS LR5

Friday 14:00 ARTS LR2

Contact me G.R.Davies@bham.ac.uk

Worksheets [40%]

Exam [60%]



Inference - Introduction

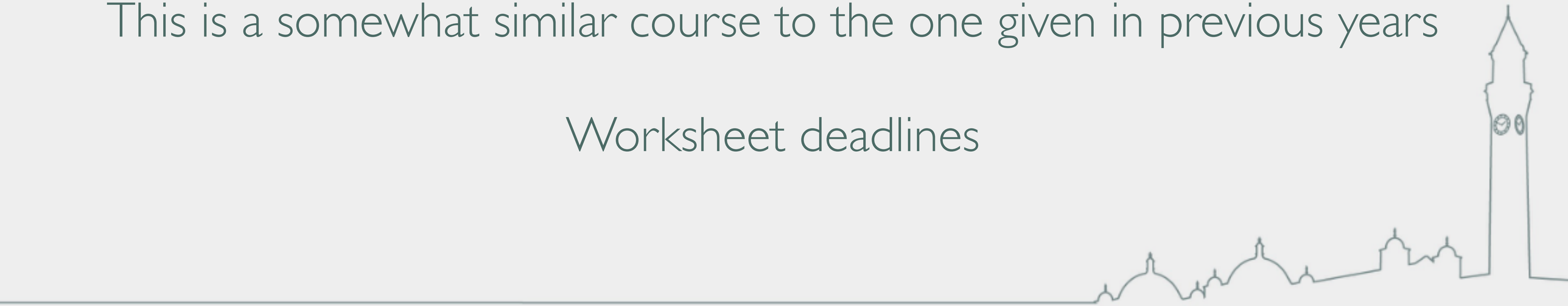
Canvas Page:

Thirdtime for me giving this course - some things will change

Canvas page will be added to as we go

This is a somewhat similar course to the one given in previous years

Worksheet deadlines



Inference - Worksheet deadlines

- Worksheet 1. Released end of week 3, due end of week 5, Friday 30th October @ 17:00:00
- Worksheet 2. Released start of week 7, due end of week 8, Friday 20th November @ 17:00:00
- Worksheet 3. Released end of week 9, due end of week 11, Friday 11th December @ 17:00:00

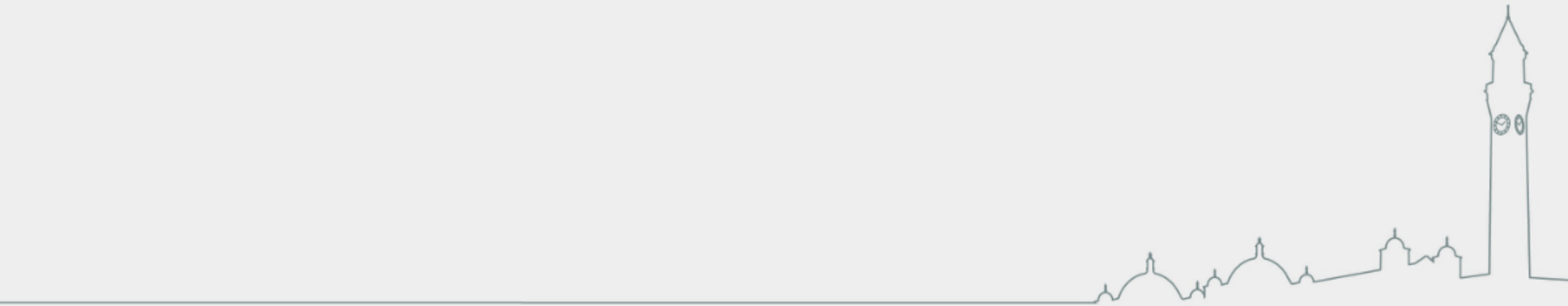


Inference - Introduction

Attendance codes:

I will put attendance codes on the board

Remind me if I forget!



Inference - Introduction

Lecture notes:

Extensive notes available on Canvas

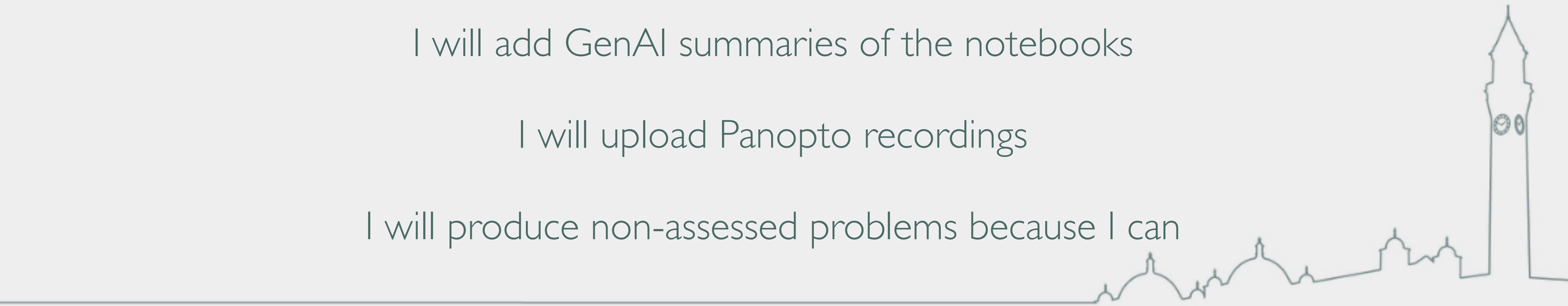
I will add slides from lectures to highlight importance

I will add Notebooks because I can

I will add GenAI summaries of the notebooks

I will upload Panopto recordings

I will produce non-assessed problems because I can



Inference - Introduction

Suggested reading list:

I should think this course is quiet self contained

If you do want to follow along using other texts …

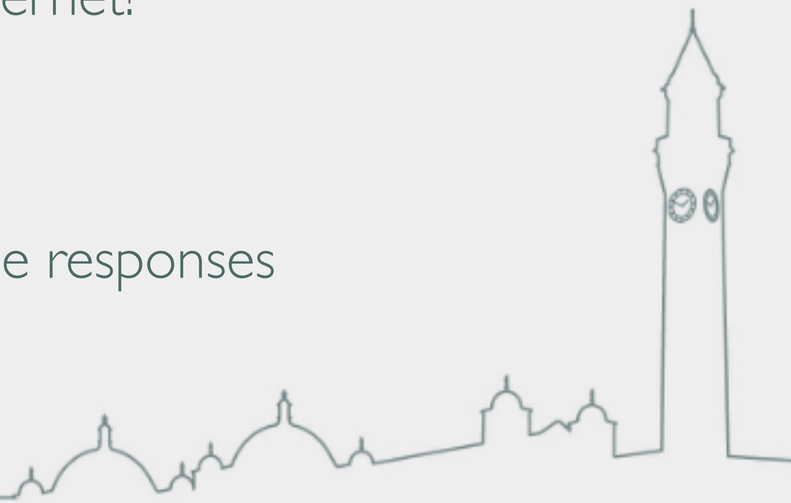
Ask me - I have opinions

There is a good body of inference work on the internet!

Note, there is also some dross

There is a reasonable amount of value in GenAI inference responses

Note, ChatGPT can make mistakes



Inference - Introduction

- **Introduction** - Intro, Logic, Probability recap
- **Probability** - Distributions, Moments, Central Limit Theorem
- **Bayesian methods** - Parameter estimation, Model selection, Stochastic sampling
- **Frequentist methods** - Model fitting, Hypothesis testing
- **'Fun stuff'** - We'll see, but maybe Gaussian Processes, …



Inference - Introduction

Today/Thursday:

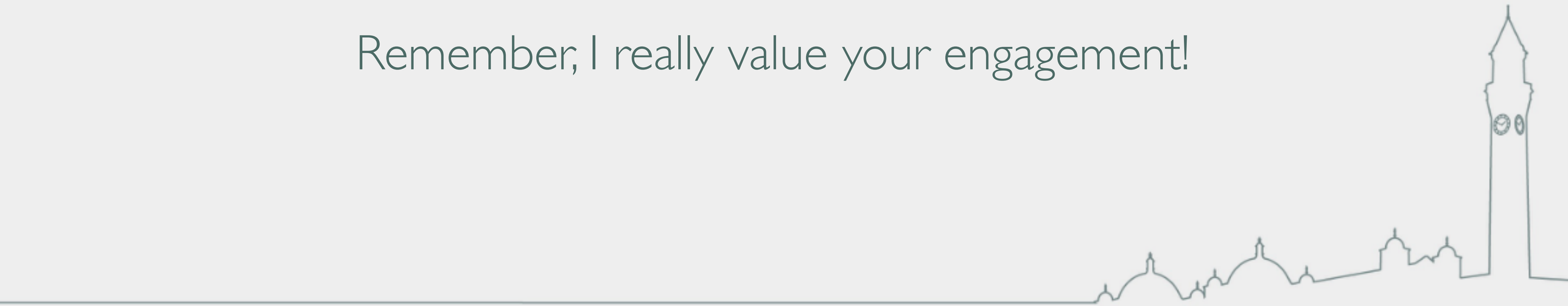
- Stats vs probability vs inference
- Logic
- Probability
- Conditional probability
- Bayes Theorem
- A Bayesian Example
- Are probabilities objective?



Inference - Introduction

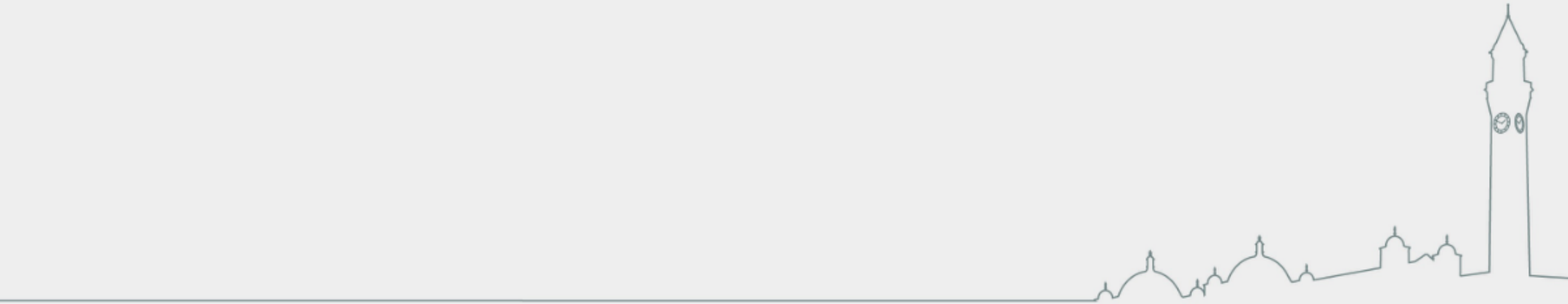
Any questions ?

Remember, I really value your engagement!



Inference - Introduction

Statistics / Probability / Inference



Inference - Logic

Events and logic

Events will use capital letters and will typically be A, B, C. Logic can be thought of as a precusor to probability.

Negation

$\bar{A}$ (which is not A)

And

$(A \text{ and } \mathrm{B}) = \mathrm{C}$

$(A, B) = C$

Or

$(A \text{ or } \mathrm{B}) = \mathrm{C}$

By convention here defined to be inclusive OR.



Inference - Logic

Idempotence

An operation that can be performed multiple times without changing the result.

$$A, A = A$$

$$(A \text{ or } \mathrm{A}) = \mathrm{A}$$

Associativity

Associative operations are important because they are precisely those operations that link objects together into a sequence.

$$A, (B, C) = A \text{ and } (\mathrm{B} \text{ and } \mathrm{C})$$

$$(A, B), C = A, (B, C) = A, B, C$$



Inference - Probability

Probability - events are now probable

$P(A)$ - The probability of event A.

The probability of event A is zero or greater but less than or equal to 1. This is by convention.

The event A could also be binary, true or false, with probability of 0 or 1 with some convention.

Logic to probability

All of the rules of logic can be made into probabilities. E.g.,

$$P(A, A) = P(A)$$

$$P((A, B), C) = P(A, B, C)$$



Inference - Probability

The law of total probability

For a set of events that are exhaustive (at least one is true) and mutually exclusive (not possible for more than one to be true),

$$A_1, A_2, \ldots, A_N$$

then

$$\sum_{i=1}^{N} P(A_i) = 1$$



Inference - Probability

Conditional Probability

The product rule

$$P(A, B) = P(A|B)\,P(B)$$

where $P(A|B)$ reads as the probability of A given B.

Symmetry of the and operator (Commutativity) ¶

$$P(A, B) = P(B, A)$$

We can see this from a Venn diagram.



Inference - Probability

The law of total probability

For a set of events that are exhaustive (at least one is true) and mutually exclusive (not possible for more than one to be true),

$$A_1, A_2, \ldots, A_N$$

then

$$\sum_{i=1}^{N} P(A_i) = 1$$

and if we want the probability of event B that is conditional on A,

$$P(B) = \sum_{i} P(B|A_i)\, P(A_i)$$



Inference - Probability

Bayes theorem

We now have enough information to derive Bayes theorem:

From the product rule ...

$$P(A, B) = P(A|B)\,P(B)$$

$$P(B, A) = P(B|A)\,P(A)$$

and from symmetry

$$P(A, B) = P(B, A)$$

and so

$$P(A|B)\,P(B) = P(B|A)\,P(A)$$

or in the common form

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$



Inference - Probability

Bayesian Example

Let's go through a common example used where you are testing to see if you are sick. The test returns either a positive you are sick or a negative you are not sick. And, of course, you are either sick or not sick. We start with Bayes Theorem:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)},$$

which becomes ...

$$P(\text{sick}|\text{tested positive}) = \frac{P(\text{tested positive}|\text{sick})\,P(\text{sick})}{P(\text{tested positive})}.$$

In this simple case, where we can consider all possible ways to testing positive on a sensible timescale, we will use the law of total probability:

$$P(B) = \sum_i P(B|A_i)\,P(A_i),$$

or

$$P(\text{tested positive}) = P(\text{tested positive}|\text{sick})\,P(\text{sick}) + P(\text{tested positive}|\bar{\text{sick}})\,P(\bar{\text{sick}}).$$

Notice how we are either sick or not sick as the exhaustive and mutually exclusive set of events.



Inference - Probability

In [5]:

```
P_tested_positive_given_sick = 0.99
P_tested_positive_given_not_sick = 0.001
P_sick = 1e-5
P_not_sick = 1 - P_sick

P_tested_positive = P_tested_positive_given_sick * P_sick + P_tested_positive_given_not_sick * P_not_sick

print(f'Probability tested positive = {P_tested_positive}')

P_sick_given_tested_positive = P_tested_positive_given_sick * P_sick / P_tested_positive

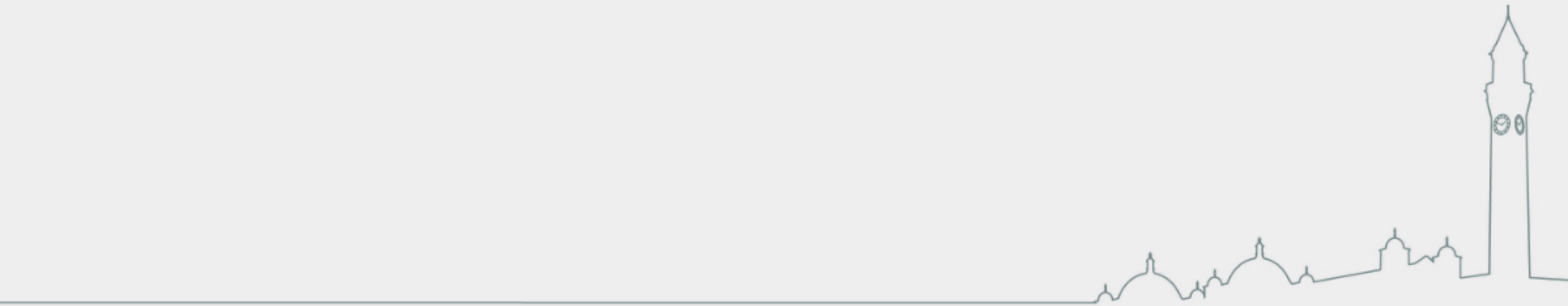
print(f'Probability I am sick given I tested positive = {P_sick_given_tested_positive}')
```

```
Probability tested positive = 0.00100989
Probability I am sick given I tested positive = 0.009803047856697264
```



Inference - Probability

Are probabilities objective?



Inference - Next time …

Probability distribution definitions

PMF/PDF

Moments

Moment generating functions

Distribution examples

Quiet a lot of maths (sorry)

